TALIS SERIES · RACK-MOUNT LFP BATTERY

Talis 5

51.2 V · 100 Ah · 5.12 kWh $LiFePO_4$ Battery Module

A 5.12 kWh rack-mount lithium iron phosphate battery module with a smart BMS, RS485 / CAN communication and front-access terminals. Designed for solar hybrid and backup systems in homes, shops, offices and small businesses.



$LiFePO_4$ 16S 1P | Smart BMS | RS485 · CAN | 19" Rack Mount | UN38.3 cells

51.2 V	100 Ah	5.12 kWh	≥2500
NOMINAL VOLTAGE	RATED CAPACITY	NOMINAL ENERGY	CYCLES @ 80% DOD

PRODUCT HIGHLIGHTS

Inverter compatible
Works with residential single- and three-phase 48 V (low-voltage) hybrid inverters.

Safe & reliable
Lithium iron phosphate (LFP) cells tested to UN38.3 — thermally stable and cobalt-free.

Easy installation
19-inch rack-mount format with front handles, front terminals and plug-in communication.

Smart BMS
Advanced BMS manages charge and discharge power and protects every cell.

Flexible expansion
5.12 kWh modular design, scalable up to 16 modules (≈ 82 kWh) in parallel.

Long life
≥ 2,500 cycles at 80% DoD, 100 A continuous charge and discharge.

MADE FOR HOMES & SMALL BUSINESSES

Homes & villas | Shops & showrooms | Small offices | Godowns & warehouses | Clinics & salons

Technical Specifications

Talis 5 · 51.2 V 100 Ah LFP Battery Module

ELECTRICAL

Nominal voltage	51.2 V
Rated capacity	100 Ah (0.5C, 25 °C)
Nominal energy	5.12 kWh
Chemistry	Lithium iron phosphate ($LiFePO_4$)
Cell & configuration	3.2 V 100 Ah · 16S 1P
Cycle life	≥ 2,500 cycles @ 1C, 80% DoD*
Monthly self-discharge	< 3%

CHARGE & DISCHARGE

Charge voltage	54.7 ± 0.2 V
Charge method	0.2C to 57.6 V, then 58.4 V to 0.02C cut-off
Standard charge current	20 A
Max. charge current	100 A
Continuous discharge current	100 A
Discharge cut-off voltage	44 V
Charge efficiency	100% @ 0.2C
Discharge efficiency	96 – 99% @ 1C

COMMUNICATION & SYSTEM

Communication	RS485 / CAN (Modbus for parallel operation)
Compatible inverters	Growatt, Deye, GoodWe, Sofar and other 48 V (LV) inverters
Scalability	Up to 16 modules in parallel (≈ 82 kWh)

MECHANICAL

Enclosure	Aluminium + GF nylon
Dimensions (D × W × H)	480 × 482 × 137 mm
Weight	≈ 38.5 kg
Installation	19-inch rack mount
Environment	Indoor

ENVIRONMENTAL

Operating temperature	0 to 55 °C (32 to 131 °F)
Relative humidity	5 – 95%, non-condensing

PROTECTION

BMS protections	Over / under-voltage, over-charge and over-discharge current, charge-current limiting, short circuit, overload

DIMENSIONS



FRONT PANEL

Power terminals	Positive / negative DC terminals with insulating covers
DC breaker	Module isolation breaker
RS485 / CAN ×3	Two link ports for parallel modules, one port to the inverter
RUN · ALARM · SOC	Status LEDs and 4-bar state-of-charge indicator
Reset	BMS reset / wake-up button
Handles	Front handles and 19-inch rack-mount ears

NOTES

- Nominal energy = nominal voltage × rated capacity.
- *Cycle life subject to the Touch Energy commercial warranty terms and operating conditions.
- Use only with a compatible inverter / charger and within the specified operating limits. Install indoors.
- Touch Energy reserves the right to change specifications without prior notice.